DENNIS G. PEZAN

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EDUCATION

Bachelor of Science in Operations Research, **United States Military Academy** Expected 2029

- Minor in Artificial Intelligence; Finance Specialization; CQPA: 3.571
- Research Project: Threat-Adaptive Interdiction of Taiwan's Crude Oil Supply (Advisor: LTC Matthew Webb)
- Conducted AIAD at the Transformation Decision Analysis Center (2026)

RESEARCH EXPERIENCE

Undergraduate Researcher, Department of Mathematical Sciences, USMA 2026 - Present

- Project: Threat-Adaptive Interdiction of Taiwan's Crude Oil Supply, advised by LTC Matthew Webb; extends his MIT Center for Transportation & Logistics doctoral research on reinforcement learning for maximum-flow network interdiction
- Investigates how to optimally attack and defend military logistics networks under adversarial interdiction, applying network optimization, game theory, and Markov decision processes
- Built a 57-node, 79-edge open-source model of Taiwan's crude oil import network (four sources, major straits and canals, three ports) with sourced capacities
- Designed a fixed-budget adversary action set of seven action types, each with a cost and success probability anchored on historical events
- Connected the network to a reinforcement-learning interdiction environment and verified it against an independent max-flow calculation; identified and fixed a targeting-filter issue that could hide the Strait of Hormuz

Research Intern, Modeling & Analysis Division, Transformation Decision Analysis Center, Fort Lee, VA 2026

- Academic Individual Advanced Development (AIAD) summer internship under MAJ Erica Mikelson
- Redesigned the reward function for a proximal policy optimization (PPO) reinforcement-learning agent that allocates resupply in Logistics Battle Command, the Army's stochastic sustainment simulation; trained and evaluated the agent in Python (PyTorch)
- Built two diagnostic tools that exposed four reward flaws, including an agent that outscored every baseline while starving the priority unit during engagements
- Retrained agent more than tripled the share of requests filled and outperformed a hand-coded priority-aware baseline across 101 random seeds
- Presented findings to Branch Chief LTC Ryan Miller at Fort Lee and, by call, to TDAC's AI and reinforcement-learning researchers (LTC Joseph Pedersen) at the Naval Postgraduate School in Monterey, CA

Independent Study in Machine Learning, Department of Mathematical Sciences, USMA 2025 - 2026

- Designed and completed an independent study of decision trees, bagging, and random forests (MA289) under LTC Patrick Kuiper, and presented the findings to officers
- Built an interactive random forest research site with a live classifier demonstration and hyperparameter explorer

PRESENTATIONS

1. **Pezan D** (2026). Reward Design for Reinforcement-Learning Resupply in Logistics Battle Command. *Final Results Brief to LTC Ryan Miller, Branch Chief, Transformation Decision Analysis Center.* Fort Lee, VA, July 22.
2. **Pezan D** (2026). Reward Design for Reinforcement-Learning Resupply in Logistics Battle Command. *Transformation Decision Analysis Center AI/RL Researchers (LTC Joseph Pedersen), Naval Postgraduate School.* Monterey, CA (virtual).

PROFESSIONAL DEVELOPMENT

Sea to Shining Sea Domestic AIAD, Defense Technology and Innovation 2026

- Held in-depth discussions with leaders at Palantir, Stanford's Gordian Knot Center (Col. (Ret.) Joseph Felter, Jack Faherty), OpenAI, Apple, and Meta Reality Labs (CTO Andrew Bosworth) on how defense technology is developed, procured, and fielded
- Topics included the barriers startups face in getting new technology to Army units, agentic AI and large language models for defense, supply-chain risk from geopolitical conflict, and the regulation of AI in national security
- Observed how the 4th Infantry Division employs and gives feedback on these systems under Next Generation Command and Control; visited U.S. Northern Command, NORAD, and the Cheyenne Mountain Complex

PROJECTS

Tree-Based Methods Research Site, dennis-pezan.com/ma289.html 2026

- Trained and evaluated decision trees, bagging, and random forests on the Titanic dataset in Python (scikit-learn) on an identical 80/20 split
- Computed pairwise tree correlation, overfitting gaps, and feature importance to show how ensemble decorrelation reduces variance; deployed via GitHub and Vercel

CFT Packing List, dennis-pezan.com/cft.html 2026

- Web tool that helps cadets organize and track gear for Cadet Field Training; deployed on Vercel with automatic deployment from GitHub

HONORS AND AWARDS

Dean's List, United States Military Academy Fall 2025, Spring 2026

Stokes Writing Fellow, United States Military Academy

- Endowed fellowship funding research travel, expert mentorship, and publication of original research

Sea to Shining Sea Fellow, United States Military Academy 2026

- Cross-disciplinary fellowship engaging cadets with senior leaders in national security, government, finance, and technology

PROFESSIONAL CERTIFICATIONS

Software Engineering Job Simulation, JPMorgan Chase & Co. (Forage) 2026

Modern Data Architecture with Microsoft Fabric, Microsoft (Coursera) 2026

PROFESSIONAL EXPERIENCE

Intern, GrayStone AI Solutions, Remote 2024

- Built Python and Excel tools to support financial planning and analysis workflows

Department Manager, McDonald's, Plainfield, IL 2022 - 2024

- Managed weekly scheduling for 80+ employees with 95% on-time accuracy and oversaw $20,000+ in weekly supply procurement
- Owned the store's labor metric, holding it at or below 23.5% month over month
- Promoted at age 16 as the youngest department manager in the region; sustained 90%+ customer satisfaction scores

LEADERSHIP AND VOLUNTEER SERVICE

Cadet-in-Charge, MA103 Tutoring, Center for Enhanced Performance, USMA 2026 - Present

- Lead the full tutor staff for MA103, a core mathematics course with 1,010 cadets enrolled; direct tutoring coverage for every major graded event
- Certified tutor in mathematics, chemistry, and psychology since 2025

Writing Consultant, Mounger Writing Center, USMA

– Selected as a Stokes Writing Fellow to coach cadets on strengthening the rhetoric and argument of their writing

Member, CLD-STEM Outreach Program, USMA 2025 - Present

– Bring hands-on STEM projects to students, reaching 800+ students across the Chicagoland and Greater Los Angeles areas

Cadet Ambassador, Cadet Ambassador Program, USMA 2025 - Present

– Represent West Point in outreach to prospective cadets and help foster a welcoming environment for future applicants

Member, Finance Club, USMA 2025 - Present

– Engage with industry executives from companies including Palantir, Uber, and Meta

Volunteer Senior Publicist, Recycle & Reboot, Plainfield, IL 2024

– Led outreach to secure donated technology; helped refurbish and distribute 100+ computers to underserved communities

Candidate, Illinois Premier Boys State 2024

– Elected Alderman and Township Supervisor; advanced to the Lieutenant Governor primary through competitive public speaking

PROGRAMMING LANGUAGES AND SKILLS

Programming: Python (PyTorch, scikit-learn), SQL, Java
Methods: Reinforcement learning, network optimization, machine learning
Spanish (fluent) ⋄ Security Clearance: Secret